

To further build optimised designs, this OS lets one calculate the power consumption of the system and figure out the power-loss graph. Relay nodes in the wireless network, which simply transmit information from one node to the next, are monitored by ContikiMAC radio duty cycling mechanism that wakes these up when duty calls. The developers fondly call these relay nodes as sleepy routers.

## Beat the memory hassle

The biggest problem as one moves into smaller and smaller devices is fitting the required amount of memory into the device. Managing access between the available locations is not easy either. This entire process seems to be made a lot easier with Contiki, owing to its efficient memory-handling mechanism.

Requiring less than 10kB random-access memory (RAM) and 30kB read-only memory (ROM), there are specific instructions for block allocations, and it also uses a mixture of event-driven and multi-threaded programming mechanisms. This concept, called protothreads, allows saving memory, while providing a neat code-control flow. The kernel, when triggered by signals like internal timers or external sensors, invokes the protothread.

**As refreshing as coffee.** One might wonder how coffee comes into the picture in an article about an OS. Well, that is what the developers call their lightweight flash file

## Contiki titbits

- Contiki is compatible with platforms like 8051-powered systems on chip, MSP430, AVR and a variety of ARM devices. The whole list can be found at [www.contiki-os.org/hardware.html](http://www.contiki-os.org/hardware.html). The available build system lets the user compile applications on the chosen platform and even try out new ones.
- Contiki can also run on computers like Apple-II series, Casio-Pocket Viewer, Intel and AMD systems, to name a few. Even gaming consoles like Atari, Jaguar and Nintendo work well with Contiki.
- Coding is completely in C language. Development and debugging can be performed via Contiki Shell built along Unix-style pipelines.
- Regression test scripts, which are used to test Cooja, can be used as a starting point for setting up simulations or to investigate how different Contiki mechanisms work.
- With a team of hardcore developers from Atmel, Cisco, ETH, Redwire LLC, SAP, Thinsquare and many others, this open source IoT OS has a strong support base to guide those new to Contiki.
- Along with Contiki package, one needs VMWare Player to make it work. The entire package of Contiki can be downloaded at a single stroke using Instant Contiki.

system! Just like a cup of coffee to ease tension, Coffee lets application programs seamlessly access external flash memory, without having to worry about either clearing flash before use or flash wear-levelling.

## Providing advanced capabilities

The environment of Contiki is built to assist designers develop the perfect code. The basics are simple with coding in standard C language. Cooja Network Simulator, the simulation environment for Contiki, lets the designer visualise a large-scale network operation or even emulate it on hardware. A dynamic module-loading feature allows one to modify designs even during run-time, easing the debug process.

In addition to features like multi-threading and protothreads, Contiki allows for multi-tasking and

inter-process communication using messages passing through events. The user can also choose to work with a graphical user interface via direct graphic support or networked remote display with virtual network computing. The OS also claims to support the world's smallest Web browser.

As Wikipedia puts it, "Contiki is an OS for networked, memory-constrained systems with a focus on low-power wireless IoT devices." Specifications of a typical Contiki system are a few kilobytes of memory, milli-watt power consumption, processing speed in the order of megahertz and a bandwidth the order of about hundreds of kilobytes. With such features and specifications, the OS seems to promise substantial work in the coming years. **EFY**

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